

Large scale dataset*



ARC-Trace: Reasoning Supervision at Scale

Auto labeling demonstrations with reasoning traces that link action chunks to its cause and intended effect.



Keyframes

External camera

Action chunk

Wrist camera

$$\begin{bmatrix} 0.155, 0.842, \dots 0.000 \\ 0.156, 0.849, \dots 0.000 \\ 0.179, 0.916, \dots 0.000 \end{bmatrix}$$

(15, 8)

Action-grounded causal reasoning traces

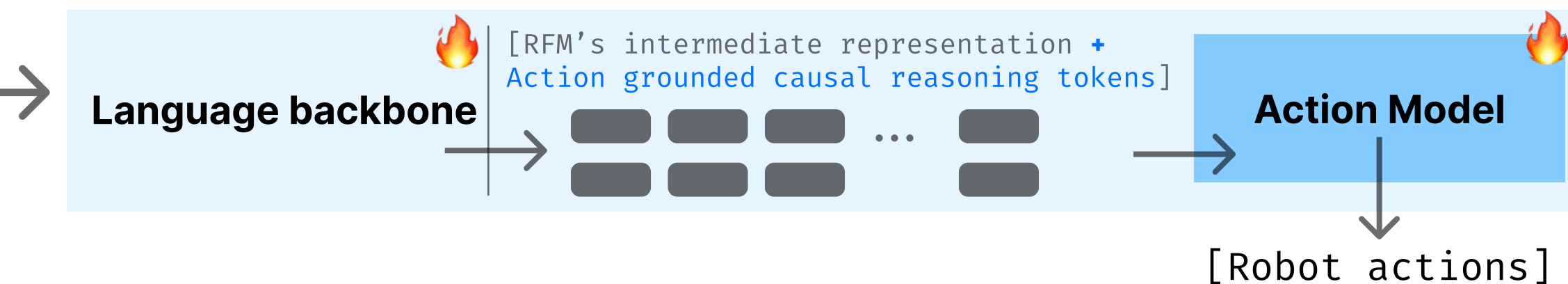
“**[State]** The robot sees the green can... in the left basin... **[Cause]** ... can must be held securely before it can be lifted... **[Effect]** The chosen effect is a stable grasp on the can... **[Action]** ... implies the robot should close its gripper around... **[Avoid]** ... robot must not push the can against the sink wall...or begin the transfer while the can is insecurely held... **[Completion]** The task is incomplete, since... can is still...”



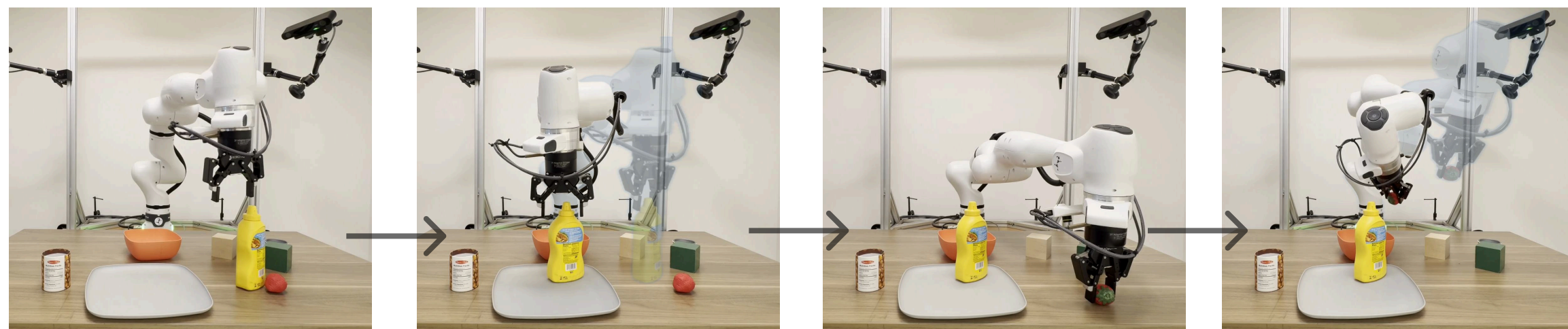
ARC Fine-tuning Recipe



Finetuned using ARC recipe



Example task: “Move the mustard bottle and place whatever is behind it into the bowl”



“... need to move the mustard first, so... should grasp it securely and move it aside...”

“...The mustard is now on the plate. The strawberry was behind it, so that is the object to put in the bowl...”

“... the gripper is around the strawberry... need a secure grasp at the center before lifting it toward...”

“...The strawberry is approaching the bowl... center it over the opening, then lower inside...”

ARC-fine-tuned models lead across benchmarks, remain robust to instruction variations, and outperform baselines on challenging reasoning-intensive tasks.



Base ARC-finetuned variant
RoboLab-120

Reasoning-50

MolmoSpaces

